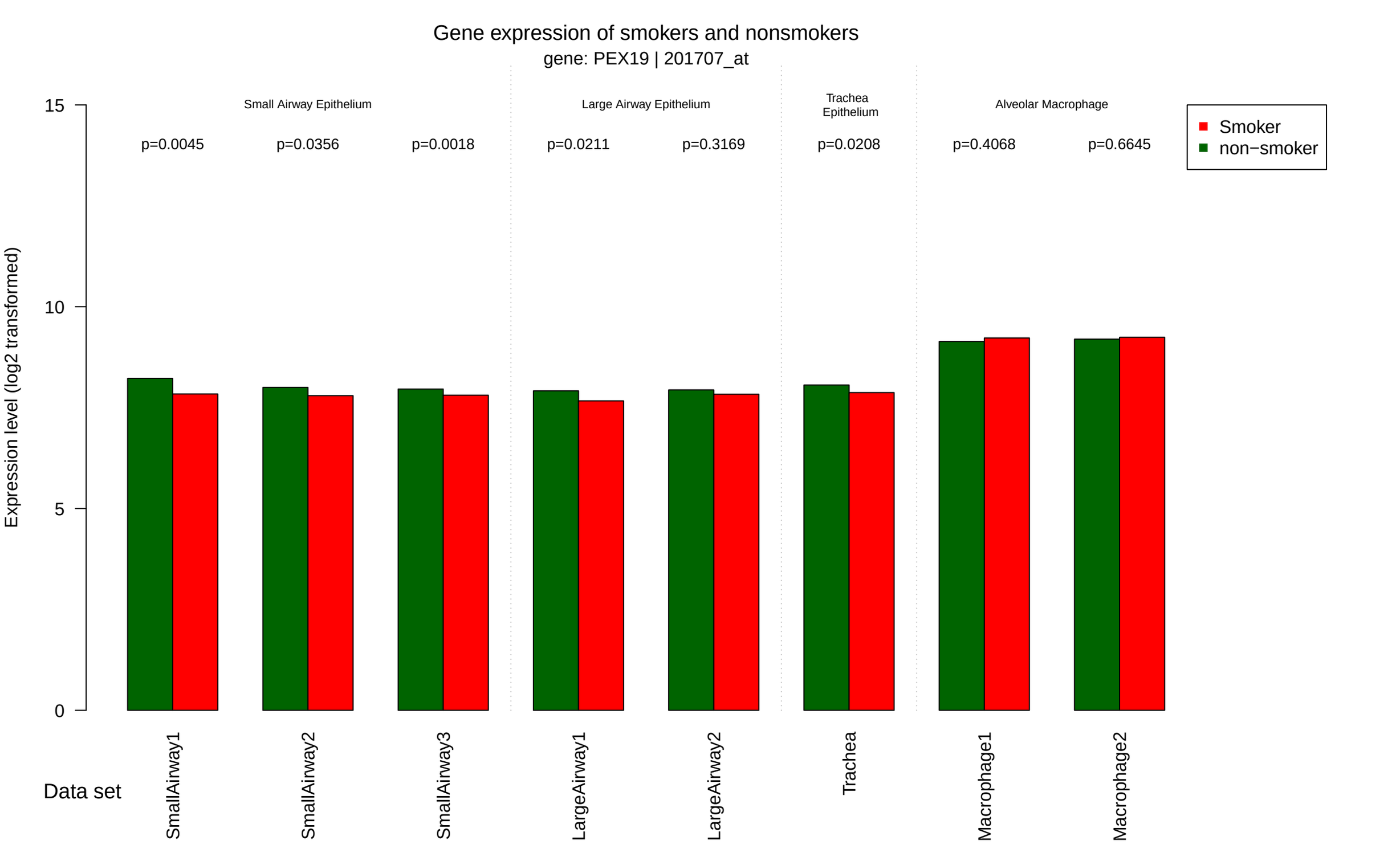
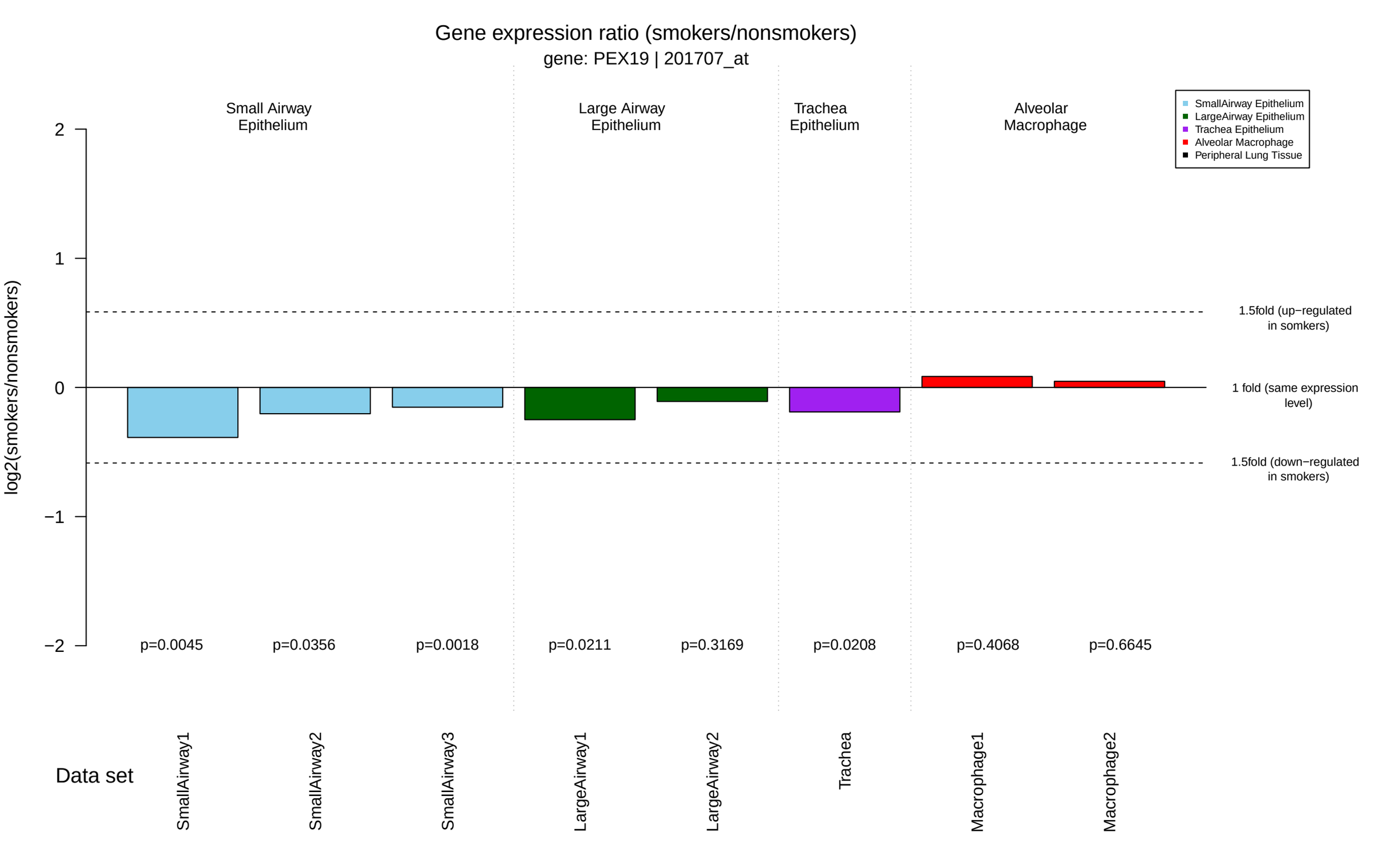
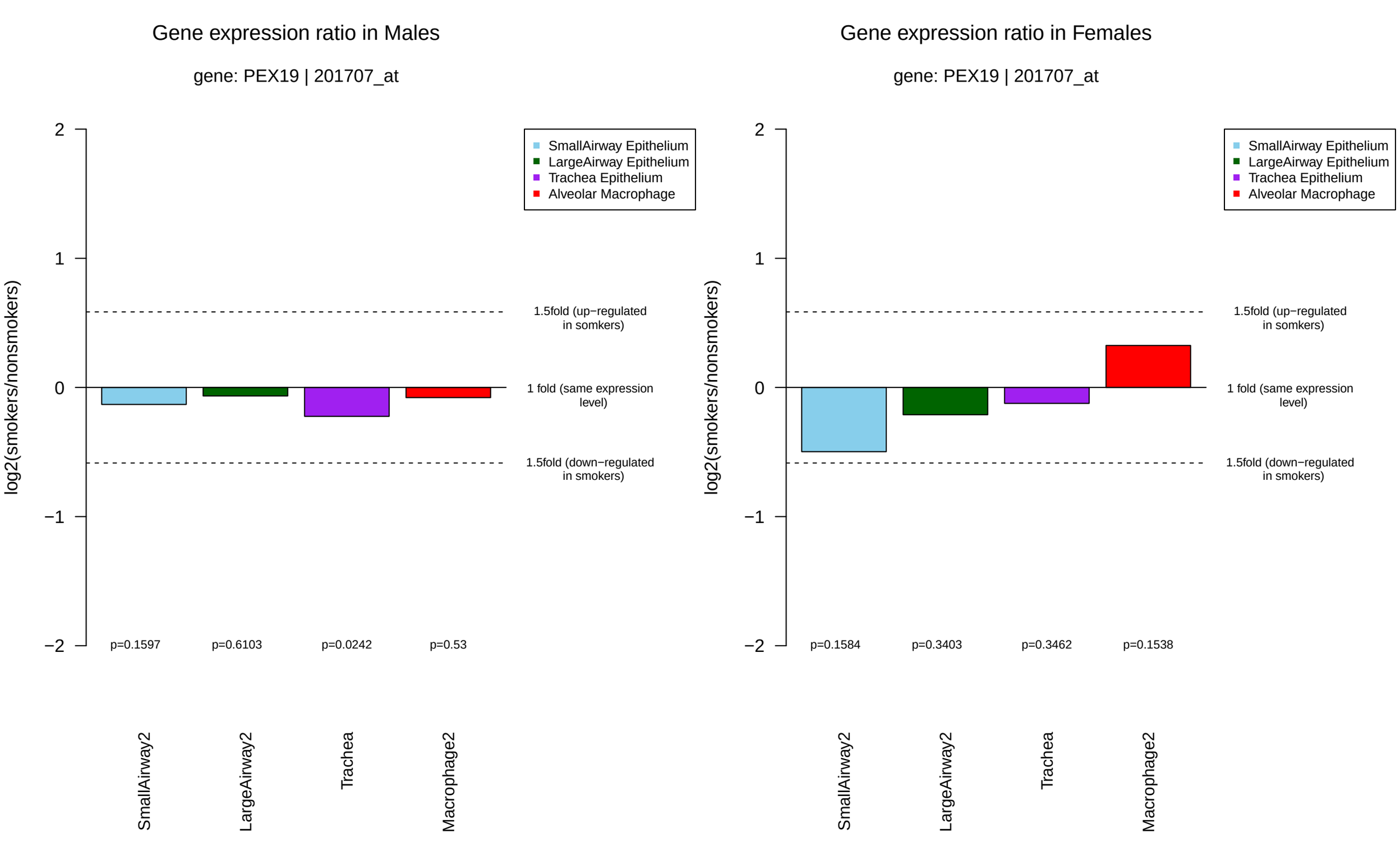
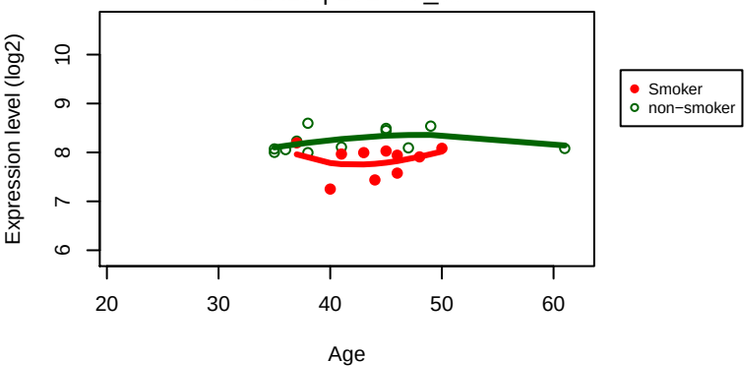
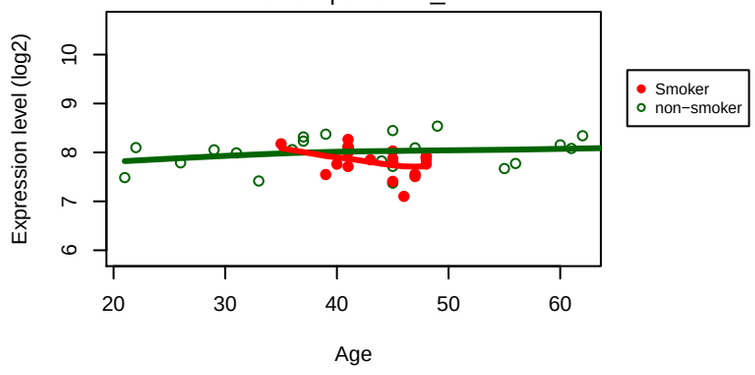
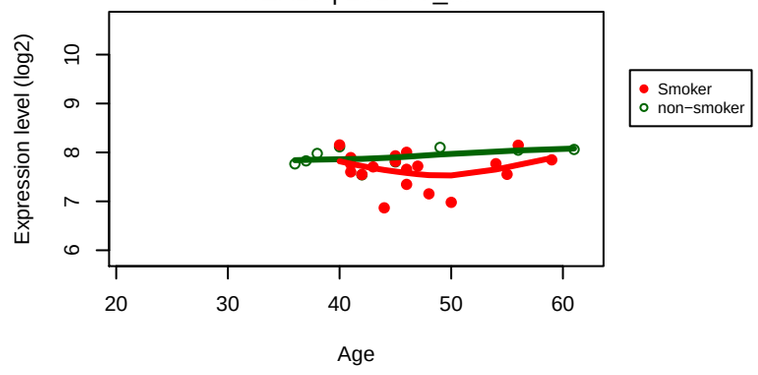

Gene expression of smokers and nonsmokers
gene: PEX19 | 201707_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0045
p=0.0356
p=0.0018
p=0.0211
p=0.3169
p=0.0208
p=0.4068
p=0.6645
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

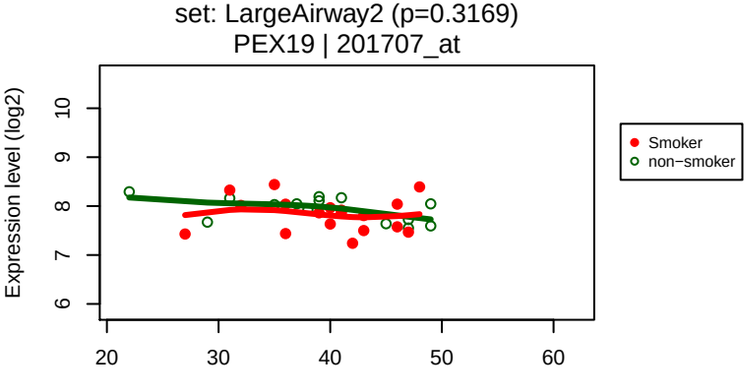

Gene expression ratio (smokers/nonsmokers)
gene: PEX19 | 201707_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0045
p=0.0356
p=0.0018
p=0.0211
p=0.3169
p=0.0208
p=0.4068
p=0.6645
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

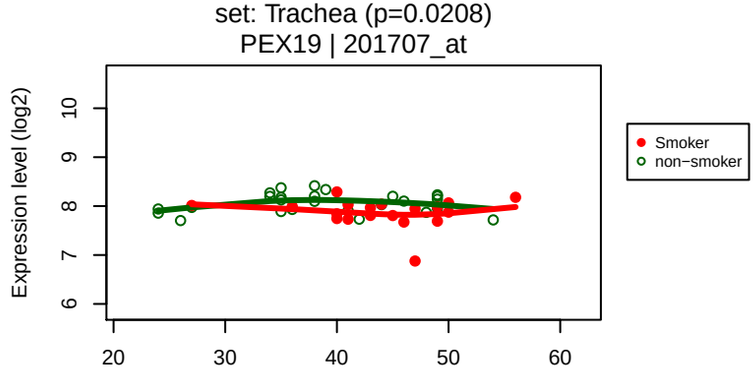

Gene expression ratio in Males
gene: PEX19 | 201707_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1597
p=0.6103
p=0.0242
p=0.53
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PEX19 | 201707_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1584
p=0.3403
p=0.3462
p=0.1538
SmallAirway2
LargeAirway2
Trachea
Macrophage2

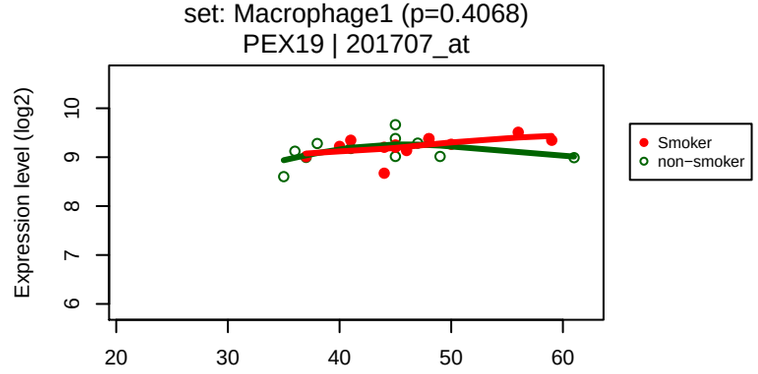

set: SmallAirway1 (p=0.0045)
PEX19 | 201707_at
Expression level (log2)
Age
Smoker
non−smoker

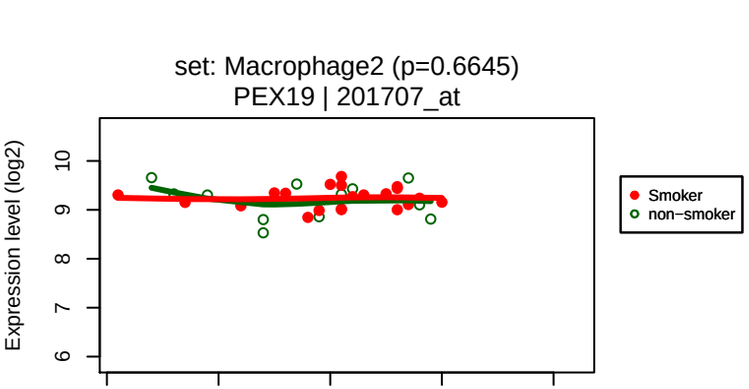

set: SmallAirway2 (p=0.0356)
PEX19 | 201707_at
Expression level (log2)
Age
Smoker
non−smoker

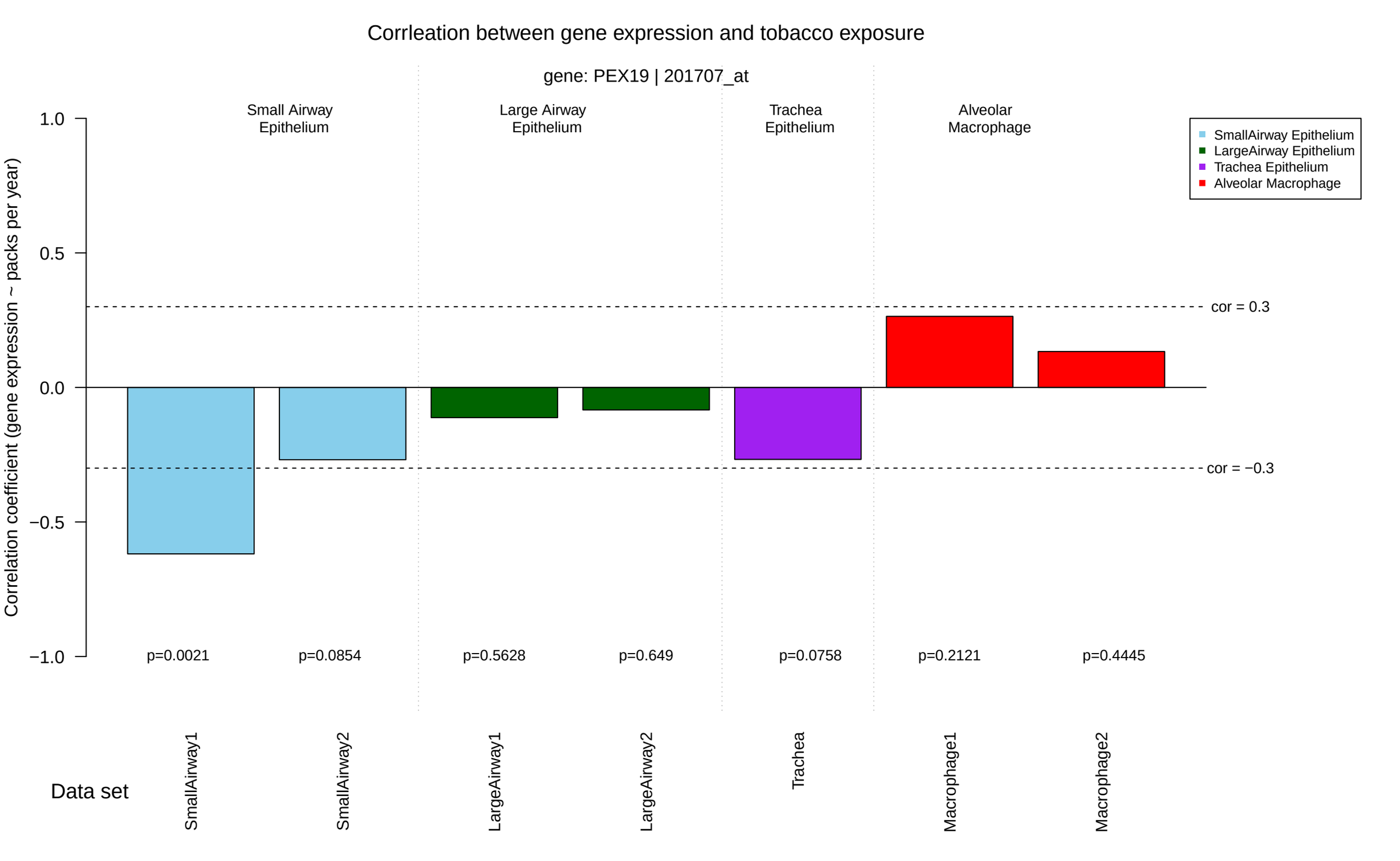

set: LargeAirway1 (p=0.0211)
PEX19 | 201707_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3169)
PEX19 | 201707_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0208)
PEX19 | 201707_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4068)
PEX19 | 201707_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6645)
PEX19 | 201707_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PEX19 | 201707_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0021
p=0.0854
p=0.5628
p=0.649
p=0.0758
p=0.2121
p=0.4445
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2